Closure slope study: nothing above the anchors resolved; degree 5 at 40 unknowns is beyond this container

These are the results of `research/dreg_closure_slope_20261018/`. The full text is `RESULTS.md`, and the run report is `reports/run-20261018T0618Z/`. Commit `9a44015c1`. **Class:** stage diagnostic. The "distance remaining" rows are **derived** from the cost model, not measured.

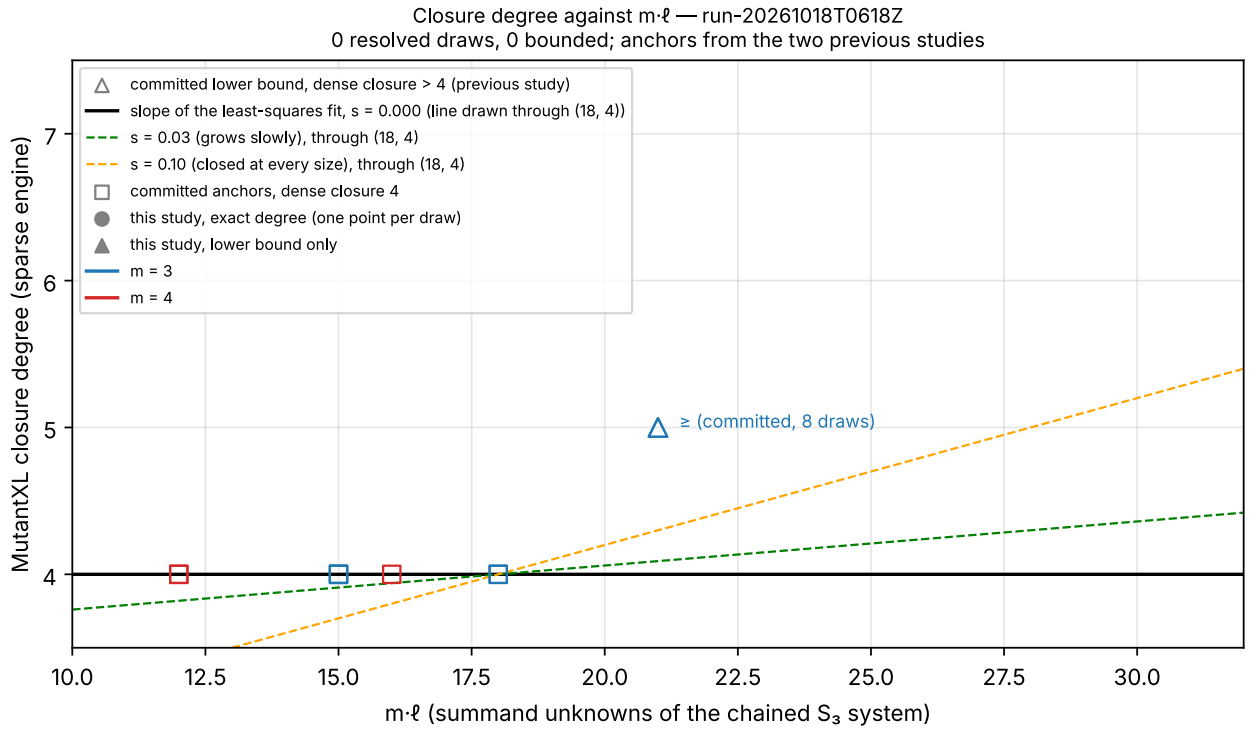


Figure 1: Closure degree per draw against $m \cdot \ell$, run run-20261018T0618Z.

Registered outcomes

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Slope s = 0.0000 degrees per summand unknown (m·ℓ), least squares over 8 cells (8 anchors at degree
4, 0 measured here)
Registered verdict: nothing resolved above the anchors: undecided
P1 (every resolved draw at m·ℓ = 20, 21 reads 5): 0 of 0 → unmeasured
P2 (a resolved cell with m·ℓ ≤ 30 reads 6): refuted on the resolved cells (or unmeasured)
P3 (s ≥ 0.10): unmeasured
P4 (sparse degree-4 replays of committed dense refutations refute): 1 of 1 measured (2 runs) →
incomplete (capped or killed runs are unmeasured)
P5 (no satisfiable draw refuted): 0 control runs, 0 refuted, 0 unmeasured
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Per-cell degrees (measured)

m	(n, ℓ)	m·ℓ	N	degree per draw	cell	secs per draw
4	(12, 3)	12	36	4 4 4 4 (committed, dense)	resolved at 4	—
4	(14, 3)	12	40	4 4 4 4 (committed, dense)	resolved at 4	—
3	(11, 5)	15	26	4 4 4 4 (committed, dense)	resolved at 4	—
3	(13, 5)	15	28	4 4 4 4 (committed, dense)	resolved at 4	—
3	(15, 5)	15	30	4 4 4 4 (committed, dense)	resolved at 4	—
4	(13, 4)	16	42	4 4 4 4 (committed, dense)	resolved at 4	—
3	(16, 6)	18	34	4 4 4 4 (committed, dense)	resolved at 4	—
3	(18, 6)	18	36	4 4 4 4 (committed, dense)	resolved at 4	—
3	(19, 7)	21	40	unmeasured unmeasured unmeasured unmeasured	unmeasured	277 269 247 264
3	(21, 7)	21	42	unmeasured unmeasured unmeasured unmeasured	unmeasured	220 223 220 231
3	(22, 8)	24	46	unmeasured unmeasured unmeasured	unmeasured	8548 0 0

Prediction	outcome	confidence	evidence
P1: every resolved draw at m·ℓ = 20, 21 reads 5	**unmeasured**: no draw resolved (all capped at degrees 5 and 6)	about 65%	`score-output.txt`
P2: a resolved cell with m·ℓ ≤ 30 reads 6	**unmeasured**: no cell above the anchors resolved	about 55%	same
P3: s ≥ 0.10	**unmeasured**: the fit has no point above the anchors	about 70%	same
P4: sparse degree-4 replays of committed dense refutations refute	**holds on 1 of 1 measured**, 1 of 2 runs killed (worker restart): incomplete	above 95%	`runs/check-*`
P5: no satisfiable draw refuted	**not run**: every registered control sits at N ≥ 46, where degree 4 itself caps	above 95%	—

- **Registered verdict:** "nothing resolved above the anchors: undecided".
- **Descriptive reading (not registered):** the committed records fix the closure degree at 4 for m·ℓ ≤ 18 and above 4 at m·ℓ = 21, so the local slope over 18–21 is at least 0.33; a fitted slope below 0.10 would need the degree to hold at 5 to about m·ℓ = 28, below 0.03 to about 51. Both are consistent with the data and untested by it.
- **Feasibility wall (measured):** at N = 40 the sparse closure caps at 350 M entries in about 3 minutes at degrees 5 and 6; F4 (a probe, not registered) was stopped by its 1 GiB step-matrix cap and, with the cap raised, killed at 12.6 GB RSS in the degree-5 step. At N = 46 even degree 4 caps after 8,466 s.
- **Soundness control:** none run here (see P5); the sparse engine rests on its agreement test with the dense closure and on P4.
- **Scope:** every statement is about m·ℓ ≤ 24 on this 4-core, 15 GB host.

Distance remaining (derived)

Derived (cost model, w = 1; extrapolated degree 4 + s·(m·ℓ − 18) at ℓ = ⌈n/m⌉, read from the fit over m·ℓ ≤ 30):

n	m	ℓ	$\log_2$ budget per solve	admissible D (chained N)	admissible D (N = m·ℓ)	extrapolated degree	gap, degrees (chained)
73	3	25	13.3	none	2	4.0	—
73	4	19	18.3	2	3	4.0	+2.0
73	5	15	19.0	2	3	4.0	+2.0
73	6	13	21.4	2	4	4.0	+2.0
109	3	37	19.6	2	3	4.0	+2.0
109	4	28	27.6	3	5	4.0	+1.0
109	5	22	29.3	3	5	4.0	+1.0
109	6	19	33.7	4	6	4.0	+0.0
127	3	43	22.7	3	3	4.0	+1.0
127	4	32	30.7	4	5	4.0	+0.0
127	5	26	36.4	4	6	4.0	+0.0
127	6	22	39.8	5	7	4.0	−1.0
131	3	44	22.8	3	3	4.0	+1.0
131	4	33	31.8	4	5	4.0	+0.0
131	5	27	38.4	5	7	4.0	−1.0
131	6	22	37.9	4	7	4.0	+0.0
163	3	55	29.4	4	4	4.0	+0.0
163	4	41	40.4	5	7	4.0	−1.0
163	5	33	47.1	6	8	4.0	−2.0
163	6	28	52.5	6	10	4.0	−2.0
233	3	78	40.2	5	6	4.0	−1.0
233	4	59	59.2	7	10	4.0	−3.0
233	5	47	67.9	8	12	4.0	−4.0
233	6	39	72.3	8	13	4.0	−4.0

The table's "extrapolated degree" column carries the anchors' value 4 because the fit has no point above them; read against the measured lower bound (≥ 5 at $m \cdot \ell = 21$, far below the priced $m \cdot \ell \approx n$), every $n = 131$ row with admissible degree ≤ 4 is short by at least one degree at $w = 1$ and by three at $w = 2.2$. The next cell is the same one: $m \cdot \ell = 21$ at degree 5, which needs a host with 64 GB or more (dense closure about 72 GB at $N = 40$; F4′s degree-5 step above 12.6 GB) or an out-of-core closure. The cells, draws and scorer are in place for it.